BACKDROPS · SETUP

Open Scratch with a dark neon backdrop ready for the action

STEPS TO FOLLOW

1. Go to **scratch.mit.edu** → click "Create"
2. Right-click the cat sprite → Delete
3. Click "Choose a Backdrop" → search "Stars" or "Neon Tunnel"
4. (Optional) Backdrops tab → draw a bright neon horizon line
5. Sign in to save your project automatically

💡 TIP

A dark backdrop makes neon-coloured sprites pop — same trick used by Geometry Dash and Tetris.

🤔 DID YOU KNOW?

The Scratch stage is 480×360 pixels. Y goes from -180 (bottom) to +180 (top). A horizon at y=-100 leaves perfect jump room.

SPRITES · COSTUMES

Paint a simple blue cube sprite with a tiny smiling face

📜 SCRIPT REFERENCE

Design the Cube

STEPS TO FOLLOW

1. Hover "Choose a Sprite" → click **Paint** 🖌️
2. Costumes tab → pick Rectangle tool, bright blue fill, no outline
3. Drag a 50 × 50 square at the centre of the canvas
4. Brush tool → two black dots for eyes, tiny smile
5. Rename the sprite "**Cube**" in Sprite Info

💡 TIP

Keep the cube small (~50 px) so jumps look snappy and spikes feel close to the action.

🤔 DID YOU KNOW?

Geometry Dash characters started as one-colour squares — drawn in 10 minutes. Simple shapes animate beautifully.

MOTION · EVENTS

On green flag, the cube starts at the left of the screen sitting on the ground

📜 SCRIPT REFERENCE

Stand on the Ground

SCRATCH BLOCKS (CUBE SPRITE)

```
when 🏁 clicked
go to x: (-150) y: (-100)
set size to (80) %
```

💡 TIP

Always reset position in the green-flag script. Players love restarting — your game must reset cleanly.

🤔 DID YOU KNOW?

"go to x y" is teleportation — instant, no animation. It's faster than "glide" when you just need to reset.

EVENTS · LOOPS

Press SPACE → the cube rises then falls back to the ground

📜 SCRIPT REFERENCE

Simple Jump

ADD THIS NEW SCRIPT TO THE CUBE SPRITE

```
when [space▼] key pressed
if ‹(y position) < (-95)› then
  repeat (10)
    change y by (12)
  end
  repeat (10)
    change y by (-12)
  end
end
```

💡 TIP

The outer "if" stops mid-air double-jumping — "only jump if near the ground".

🤔 DID YOU KNOW?

This jump pauses the script. Step 5 upgrades it to a smarter version so spikes can still scroll while you jump.

VARIABLES · PHYSICS

Replace the simple jump with "y power" + gravity so spikes can scroll while you jump

📜 SCRIPT REFERENCE

Add Gravity

REPLACE THE SPACE SCRIPT — TWO SCRIPTS NOW

```
when 🏁 clicked
go to x: (-150) y: (-100)
set [y power▼] to (0)
forever
  change [y power▼] by (-1)
  change y by (y power)
  if ‹(y position) < (-100)› then
    set y to (-100)
    set [y power▼] to (0)
  end
end
when [space▼] key pressed
if ‹(y position) < (-95)› then
  set [y power▼] to (12)
```

💡 TIP

Positive y = up. y power = 12 launches up; gravity subtracts 1 each frame so the leap arcs back down naturally.

🤔 DID YOU KNOW?

Two scripts running at once is "parallel programming". Real games use threads or async — same idea.

🔺

SPRITES · PAINT EDITOR

Paint a small red triangle obstacle sprite

STEPS TO FOLLOW

1. Hover "Choose a Sprite" → click **Paint**
2. Costumes tab → Line tool, bright red, thickness 6
3. Draw 3 lines forming an upward triangle (~40 px wide)
4. Fill tool 🪣 → fill the inside red
5. Rename the sprite "**Spike**"

💡 TIP

Anchor the spike's base at the bottom of the canvas — then "go to y: -110" plants it on the ground perfectly.

🤔 DID YOU KNOW?

Geometry Dash uses only a handful of shapes — spike, block, jump-pad — arranged in patterns. Limited art + clever rhythm = endless levels.

➡️

MOTION · LOOPS

Spike slides from right to left, then loops back to the right edge endlessly

📜 SCRIPT REFERENCE

Spikes Scroll Left

SCRATCH BLOCKS (SPIKE SPRITE)

```
when 🏁 clicked
go to x: (240) y: (-110)
forever
    change x by (-6)
    if <(x position) < (-240)> then
        set x to (240)
```

💡 TIP

"change x by (-6)" is the scroll speed.

Bigger negatives = faster. Step 10

makes this speed grow with the score!

🤔 DID YOU KNOW?

The character never moves forward —

the world moves past it. Subway

Surfers and Mario use the exact same

trick.

SENSING · CONTROL

If the cube touches a spike, the game stops everything instantly

ADD INSIDE CUBE'S FOREVER LOOP

```
if <touching [Spike▼]?> then
  say [Game Over!] for (2) secs
  stop [all▼]
end
```

💡 TIP

"stop all" is the nuclear option — it freezes every script in every sprite. Perfect for game over.

🤔 DID YOU KNOW?

In pro games this is a "hitbox check" — CPU compares rectangles 60 times a second to decide if you live or die.

VARIABLES · LOOPS

Score climbs by 1 every second you stay alive — survival is rewarded

📜 SCRIPT REFERENCE

Score Counter

NEW SCRIPT ON CUBE SPRITE

```
when 🏁 clicked
set [Score▼] to (0)
forever
    wait (1) secs
    change [Score▼] by (1)
```

💡 TIP

Multiple "when 🏁 clicked" scripts are fine — Scratch runs them all in parallel, each handling one job.

🤔 DID YOU KNOW?

Time-based scoring is called "survival scoring". Subway Surfers, Temple Run, Fall Guys all use it.

VARIABLES · OPERATORS

Spike speeds up as Score climbs — automatic difficulty ramp

📜 SCRIPT REFERENCE

Speed Climb

REPLACE "CHANGE X BY (-6)" ON SPIKE WITH:

```
change x by ( 0 - (6 + ((Score) / (5))) )
```

→ *0 minus (6 plus Score divided by 5)*

→ *as Score grows, the value gets more negative*

→ *more negative x = faster left scroll!*

💡 TIP

Start gentle (6 + Score/5). Too easy?

Try Score/3. Too hard? Try Score/10.

Tune to taste!

🤔 DID YOU KNOW?

This is a "difficulty curve". Tetris

speeds up every 10 lines, Pac-Man

ghosts roam faster each maze. Same

trick everywhere.